

入上智能-42163

1. (1) $P(E=F, S=F, M=F, B=F)$

$$= P(S=F|E=F, M=F) \times P(B=F|M=F) \times P(M=F) \times P(E=F)$$

$$= 0.9 \times 0.9 \times 0.9 \times 0.6$$

$$= 0.4374$$

(2) $P(B=T) = P(B=T|M=T) + P(B=T|M=F)P(M=F)$

$$= 1 \times 0.1 + 0.1 \times 0.9$$

$$= 0.19$$

(3) $P(M=T|B=T) = \frac{P(M=T, B=T)}{P(B=T)} = \frac{P(B=T|M=T)P(M=T)}{P(B=T)}$

$$= \frac{1 \times 0.1}{0.19}$$

$$= \frac{1}{1.9} \approx 0.5263$$

(4) $P(M=T|E=T, S=T, B=T) = \frac{P(M=T, E=T, S=T, B=T)}{P(E=T, S=T, B=T)}$

$$= \frac{P(M=T, E=T, S=T, B=T)}{P(E=T, S=T, B=T|M=T) + P(E=T, S=T, B=T|M=F)P(M=F)}$$

$$= \frac{0.1 \times 0.9 \times 0.9 \times 0.6}{0.1 \times 1 \times 0.9 \times 0.9 \times 0.6 + 0.1 \times 0.9 \times 0.9 \times 0.6 \times 0.9}$$

$$= \frac{0.054}{0.054 + 0.04374} \approx 0.5574$$

(5) $P(E=T|M=T) = \frac{P(E=T, M=T)}{P(M=T)} = \frac{P(S=E=T, M=T)}{P(M=T)}$

$$= \frac{P(S=E=T, M=T)}{P(M=T)}$$

$$= \frac{P(S=E=T, M=T) + P(S=F, E=T, M=T)}{P(M=T)}$$

$$= \frac{P(E=T)P(M=T)P(S=T|E=T, M=T) + P(E=F)P(E=T, M=T)}{P(M=T)}$$

$$= \frac{0.4 \times 0.1 \times (1+0)}{0.1}$$

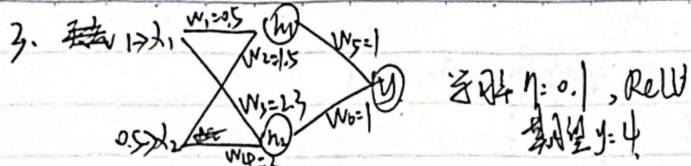
$$= 0.4$$

天气	温度	活动(进行)	熵: $info[3,2] = -\frac{2}{3}\lg\frac{2}{3} - \frac{1}{3}\lg\frac{1}{3} \approx 0.917$	总信息增益: $gain = info[3,2] - 0.917 \approx 0.042$
晴	高	No	熵: $info[1,0] = -\lg 1 = 0$	
晴	高	No	熵: $info[2,0] = -\lg 1 = 0$	
多云	高	Yes	熵: $info[0,2] = -\lg 1 = 0$	
雨	中	Yes	熵: $info[0,2] = -\lg 1 = 0$	
雨	中	Yes	熵: $info[0,2] = -\lg 1 = 0$	

总信息增益: $gain = info[3,2] - 0.917 \approx 0.042$

No.

Date



$\sum \text{Relu } \eta: 0.1$, Relu
 $\text{Relu } y: 4$

$$h_1 = 1 \times 0.5 + 0.5 \times 1.5 = 1.25, \text{Relu}(h_1) = 1.25$$

$$h_2 = 1 \times 2.3 + 0.5 \times 3 = 3.8, \text{Relu}(h_2) = 3.8$$

$$\hat{y} = w_4 \times h_1 + w_5 \times h_2 = 5.05, \text{Relu}(\hat{y}) = 5.05$$

$$\text{loss} = 5.05 - 4 = 1.05, \text{loss} = \frac{1}{2} (\hat{y} - y)^2 = \frac{1}{2} (5.05 - 4)^2$$

$$\begin{aligned}
 w_5^+ &= w_5 - \eta \times \frac{\partial \text{loss}}{\partial w_5} = w_5 - \eta \times \frac{\partial \text{loss}}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_5} \\
 &= 1 - 0.1 \times (5.05 - 4) \times h_2 \\
 &= 1 - 0.1 \times 1.05 \times 3.8 \\
 &= 1 - 0.13125 \\
 &= 0.86875
 \end{aligned}$$

$$\begin{aligned}
 w_1^+ &= w_1 - \eta \times \frac{\partial \text{loss}}{\partial w_1} = w_1 - \eta \times \frac{\partial \text{loss}}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial h_1} \times \frac{\partial h_1}{\partial w_1} \\
 &= 0.5 - 0.1 \times 1.05 \times 1 \\
 &= 0.395
 \end{aligned}$$

No. 23320093

Date 2023.10.10

4.

①

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 2 & 3 & 1 & 0 \\ 0 & 3 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

表秩数1:

$$\begin{pmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 3 \\ 3 & 3 & 5 & 1 \\ 0 & 7 & 1 & 2 \\ 0 & 3 & 2 & 1 \\ 3 & 4 & 2 & 3 \\ 2 & 2 & 6 & 2 \\ 1 & 3 & 2 & 0 \end{pmatrix}$$

增广项1 =

$$\begin{pmatrix} 2 & 6 & 6 & 2 \\ 6 & 9 & 6 & 7 \\ 6 & 6 & 12 & 4 \\ 2 & 7 & 4 & 3 \end{pmatrix}$$

②

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 & 0 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

表秩数2:

$$\begin{pmatrix} -3 & -2 & 1 & 0 & -1 & -2 & -3 & 0 \\ -2 & -2 & -1 & 1 & -2 & -2 & 1 & 3 \\ -1 & -1 & -1 & 2 & -3 & 1 & 1 & 1 \\ 0 & 1 & 3 & 2 & 0 & 3 & 1 & 2 \\ 0 & -3 & -2 & -1 & 0 & -3 & -2 & -1 \\ 3 & 0 & 0 & -3 & 2 & 0 & 0 & -2 \\ 1 & 3 & 2 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

增广项2 =

$$\begin{pmatrix} 1 & -3 & -3 & 1 \\ 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 \\ 3 & 8 & 5 & 4 \end{pmatrix}$$

③ 特征值求法: $\left| \frac{3-2+2x}{1} + 1 \right| \times \left| \frac{3-2+2x}{1} + 1 \right| \times 2.$

= $4 \times 4 \times 2.$

(2) ①. 2x2平均池化:

$$\begin{pmatrix} 5.75 & 5.25 \\ 5.25 & 5.75 \end{pmatrix}$$

$$\begin{pmatrix} 9.25 & 0.75 \\ 3.75 & 3.25 \end{pmatrix}$$

②. 2x2最大池化:

$$\begin{pmatrix} 9 & 7 \\ 7 & 12 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 3 \\ 8 & 5 \end{pmatrix}$$